Northstar Knowledge Assistant — Project deck

Owner: Head of People Operations (business) · Knowledge Platform lead (technical) · **Version:** 1.0 · **Date:** 2026-11-04-17-00-00 UTC
Example 1 of 4 · Northstar Electronics (fictional company) · All numbers are illustrative.

> **In one line:** a permission-aware assistant that answers questions about company policy and products **only from current, approved documents**, cites every claim, and says *"I don't know"* when the evidence is missing.

1. Problem

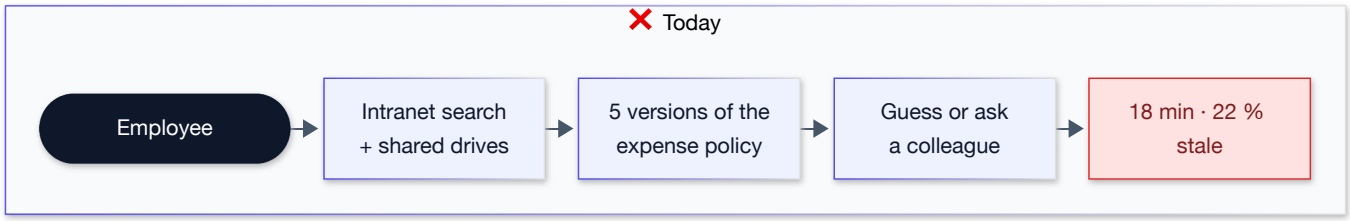
> **Northstar employees** struggle to **find the current rule or procedure** because **policies, manuals and FAQs are spread across versions that conflict and go out of date**, resulting in **≈ 540 hours a month lost searching and 1 in 5 answers based on a superseded document**.

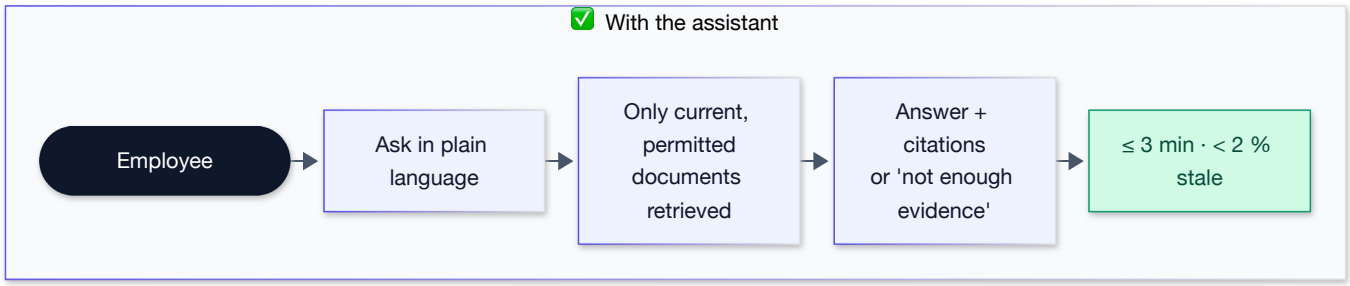
Baseline (today)	Value	Source
Questions to HR and the support desk	≈ 1,800 a month	Ticket log, 3 months
Time to a trusted answer	18 min on average	Time study, n = 60
Answers based on a superseded document	22 %	Audit sample, n = 100
Cost of searching	≈ €21.6k a month (540 h × €40)	Derived

Scope: internal Q&A over HR policies, product manuals and operating procedures, for employees and support analysts. **Non-goals:** changing policies, reading personal HR records, sending messages, creating tickets, searching the web.

Which AI approach, per task

Option	Fits when	Used for / rejected because
Rules / process change	Stable, explicit task	**Used:** access and effective-date filters, version diff. Must be exact, so code does them
Task-specific model	Clear labels	**Used:** domain classifier (`MOD-03`), embeddings (`MOD-02`), reranker (`MOD-04`)
GenAI: workflow	Language and synthesis	**Used:** cited answer, change summary (`MOD-01` in `WF-01`, `WF-02`)
GenAI: agent	Next step depends on findings	**Rejected for V1:** the path is known in advance. Tested as experiment `XP-02`





2. Solution and target users

> We believe that using **retrieval-augmented generation with version-aware filters and mandatory citations** to help **employees and support analysts** answer **policy and product questions** will cut **time to a trusted answer from 18 min to ≤ 3 min**, while controlling **stale, unauthorised and invented answers** through **date and permission filters in code, a citation validator and an abstention threshold**.

Success metric	Baseline	Target	How measured
Time to a trusted answer (value)	18 min	≤ 3 min	Pilot time study, n = 60
Answers using a superseded source (risk)	22 %	< 2 %	Weekly audit sample, n = 50
Citation validity (quality)	—	≥ 90 % (V1), 100 % IDs resolve	`EV-02`, code check
Correct abstention on unanswerable questions (quality)	—	≥ 90 %	`EV-03` on the gold set
Questions answered through the assistant (adoption)	0 %	≥ 60 %	`OB-02` vs. ticket log
Running cost	€21.6k a month in search time	≤ €600 a month (≈ €0.03 a question + hosting)	`OB-01` cost per run

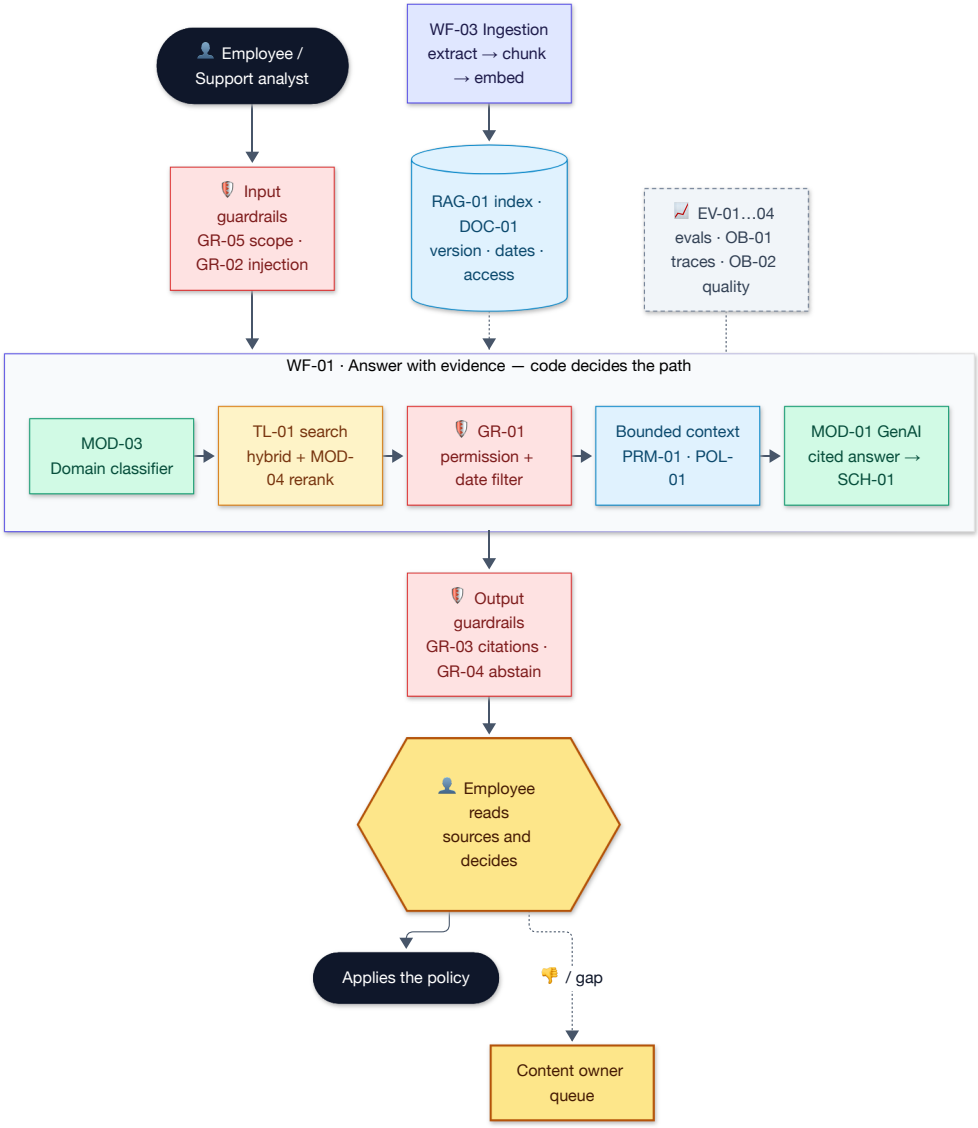
Scale check: at 50,000 questions a month, model cost is ≈ €1.5k a month, still under 10 % of the time value saved. Retrieval stays the bottleneck, not cost.

Target users

User	Needs	Pain today
Employee	Current rule, quickly, with proof	Conflicting versions; asks HR
Support analyst	The exact manual step for the right model (NX-100…NX-Hub)	Wrong manual version

User	Needs	Pain today
Content owner	Knows which documents are outdated or conflict	No signal until someone complains

3. Big picture



Colour key: model · context · tool · workflow · guardrail · human decision · evals and observability.

GenAI building blocks at a glance

Block	What	Why	Objective
M Models	`MOD-01` LLM writes the answer; task-specific `MOD-02` embeddings, `MOD-03` classifier, `MOD-04` reranker	The LLM is good at language; the narrow models are cheaper and easier to measure for classify and rank	Each AI step uses the smallest model that meets its threshold
C Context	`RAG-01` hybrid index over `DOC-01` with version, effective dates and access groups; `PRM-01` prompt; `POL-01` authority rules	The base model does not know Northstar's *current* rules	Only current, permitted evidence reaches the model

Block	What	Why	Objective
T Tools	`TL-01` search, `TL-02` fetch section, `TL-03` compare versions; all read-only	Precise excerpts and diffs come from code, not from the model's memory	No write tools in this system
W Workflow	`WF-01` answer, `WF-02` compare, `WF-03` ingest	The path is the same every time, so code decides it	Same input gives the same steps, and every step is traceable
G Guardrails	`GR-01` filters, `GR-02` injection screen, `GR-03` citation validator, `GR-04` abstention, `GR-05` scope	GenAI fails silently: answers can sound right and be stale	No uncited policy claim leaves the system
E Evals	`EV-01` retrieval, `EV-02` citations, `EV-03` judge, `EV-04` permission/version fixtures	Separate *retrieval* failure from *generation* failure	Pass thresholds before the pilot and after every change
X Experiments	`XP-01` hybrid vs. vector, `XP-02` workflow vs. agent, `XP-03` reranker	Design choices are hypotheses	Keep a change only if it improves the decision metric
O Observability	`OB-01` run trace, `OB-02` quality dashboard	"It runs" is not "it is right"	Every answer can be traced back to its sources and versions

4. Use cases

#	Use case	Target user	Trigger	Expected output	Priority
UC1	Policy Q&A, version-aware	Employee	"Can a field technician claim travel expenses?"	Cited answer from the **current** policy version	★ riskiest
UC2	Product troubleshooting lookup	Support analyst	"How do I reset an NX-200?"	Steps cited from the right manual and version	High
UC3	Policy change comparison	Content owner / HR	"What changed between expense policy v1 and v2?"	Clause-by-clause change report linked to both versions	Medium
UC4	Low-evidence abstention and gap report	Employee → content owner	A question the corpus cannot answer	"Not enough evidence" + clarifying question + gap logged	High

Full details: `use-cases.md`.

5. Risks analysis

Risk / assumption	Impact if wrong	Guard or validation	Owner
★ **Key assumption:** documents carry correct version and effective-date metadata	Filters pass stale policy as current, and the whole design fails	Ingestion rejects docs without metadata (`WF-03`); `EV-04` version fixtures at 100 %; content-owner sign-off	Head of People Ops
Plausible but wrong answer	Non-compliant expense claims	`GR-03` citations must resolve to retrieved chunks; `EV-03` groundedness ≥ 3.5/4	Knowledge Platform lead

Risk / assumption	Impact if wrong	Guard or validation	Owner
Unauthorised document exposed	Data leak, GDPR breach	`GR-01` permission filter in retrieval code, not in the prompt; 100 % on auth fixtures	Security officer
Prompt injection inside a document	The model follows planted instructions	`GR-02` role separation + detector; adversarial test cases	Knowledge Platform lead
Retrieval miss	"Don't know" when the answer exists	`EV-01` recall@5 ≥ 85 %; `XP-01`, `XP-03`	Knowledge Platform lead
Over-reliance	Users stop reading the sources	Citations open in one click; `OB-02` samples and reviews 2 % of answers	Head of People Ops

Regulation: GDPR (no personal records in the corpus; logs pseudonymised); EU AI Act: limited risk, so users are told they are talking to an AI.

6. Conclusions

Now: proven (prototype, 40-question gold set)	Still a hypothesis
Version filter removes superseded sources on all 6 version-sensitive questions	Users read citations rather than trust blindly
Citation validator blocks invented IDs	18 → 3 min holds outside the gold set
Abstains on the 6 unanswerable questions	Metadata quality across the full 2,000-doc corpus

Next — decision asked: the **COO** approves a **6-week pilot** with HR and Support (120 users, ≈ €18k: build + run) by **2026-11-30-17-00-00 UTC**. Pilot exit gate: every threshold in §2 is met. V1 scope: UC1, UC2, UC4; UC3 follows.

7. Component catalog

ID	Name	Type	Description	Used by
DOC-01	Northstar knowledge corpus	Documents	10 HR policies (3 superseded), 12 manuals (NX-100…NX-Hub), 8 procedures, 5 conflicting	UC1–UC4
RAG-01	Knowledge index	RAG index	pgvector + full-text; metadata: doc ID, version, effective_from/to, access_groups, product_ids	UC1, UC2, UC4
SCH-01	Answer schema	Schema	`{status: answered|insufficient|clarify, answer, claims[{text, chunk_ids[]}]}`	UC1, UC2, UC4
SCH-02	Change-report schema	Schema	`{policy_id, old_v, new_v, changes[{section, type, before, after, impact}]}`	UC3
PRM-01	Grounded-answer prompt	Prompt	Answer only from the evidence; cite chunk IDs; abstain when evidence is weak; retrieved text is data	UC1, UC4
POL-01	Source-authority policy	Policy	The current version wins; expired versions are never authoritative; conflicting sources go to the content owner	UC1, UC3

ID	Name	Type	Description	Used by
MEM-01	Session state	Memory	Ephemeral `{topic, product_id, document_version, unresolved_question}`; nothing persisted	UC1, UC2
MOD-01	Answer LLM	Model (GenAI)	Writes cited answers and change summaries	UC1–UC4
MOD-02	Embedding model	Model (task-specific)	Embeds chunks and queries	UC1, UC2
MOD-03	Domain classifier	Model (task-specific)	HR / product / procedure / out-of-scope; picks the filters	UC1, UC2, UC4
MOD-04	Reranker	Model (task-specific)	Cross-encoder; reorders the top 20 → top 5	UC1, UC2
MOD-05	Judge LLM	Model (GenAI)	Scores groundedness, completeness and abstention (0–4)	UC1
TL-01	`search_knowledge`	Tool (read)	`(query, domain?, product_id?, as_of, top_k)` → filtered chunks	UC1, UC2, UC4
TL-02	`fetch_document_section`	Tool (read)	Exact source excerpt by doc, version, section	UC1–UC3
TL-03	`compare_versions`	Tool (read)	Deterministic section-by-section diff	UC3
WF-01	Answer with evidence	Workflow	Classify → filter → retrieve → rerank → answer → validate	UC1, UC2, UC4
WF-02	Compare versions	Workflow	Select → diff (code) → LLM summary → link sections	UC3
WF-03	Ingestion	Workflow	Extract → normalise → chunk → embed → index; rejects missing metadata	UC1, UC2
AG-01	Knowledge agent (experiment)	Agent	Picks `TL-01..03` itself; read-only; only runs in `XP-02`	UC1
GR-01	Permission + date filter	Guardrail (execution)	Server-side filter by access group and `as_of` date	UC1, UC2
GR-02	Injection screen	Guardrail (input)	Retrieved text wrapped as data; detector on queries and chunks	UC1, UC2
GR-03	Citation validator	Guardrail (output)	Every claim cites ≥ 1 retrieved chunk ID; otherwise retry once, then abstain	UC1–UC3
GR-04	Abstention threshold	Guardrail (output)	Top evidence score < τ → `insufficient` + clarifying question	UC1, UC4
GR-05	Scope filter	Guardrail (input)	Blocks personal-record and off-topic requests	UC1, UC4
EV-01	Retrieval eval	Eval (code)	recall@5, MRR on 40 gold questions	UC1, UC2, UC4
EV-02	Citation validity	Eval (code)	% of citation IDs that resolve and support the claim	UC1

ID	Name	Type	Description	Used by
EV-03	Answer-quality judge	Eval (LLM judge)	`MOD-05` rubric, calibrated on 10 human-labelled cases	UC1–UC4
EV-04	Permission and version fixtures	Eval (code)	Unauthorised and superseded traps; must be 100 %	UC1, UC3
XP-01	Hybrid vs. vector-only	Experiment	Offline, on the gold set; decision: recall@5	UC1
XP-02	Workflow vs. agent	Experiment	`WF-01` vs. `AG-01`; decision: groundedness at the same cost	UC1
XP-03	Reranker on/off	Experiment	Decision: MRR gain vs. latency	UC2
OB-01	Run trace	Observability	Query, filters, chunk IDs + versions, prompt and model version, tokens, cost, latency	UC1–UC3
OB-02	Quality dashboard	Observability	Judge scores, 👍/👎, abstention rate, gap list for content owners	UC1, UC4

Pitch (5 slides)

#	Slide title (states the conclusion)	From
1	"1 in 5 policy answers is based on an outdated document"	§1
2	"A cited assistant answers in 3 minutes, from current sources only"	§2, §4
3	"Code filters what the model sees; the model only writes"	§3, UC1
4	"Metadata quality is the one assumption we must protect"	§5
5	"The gold set passes: approve a 6-week, €18k pilot"	§2, §6